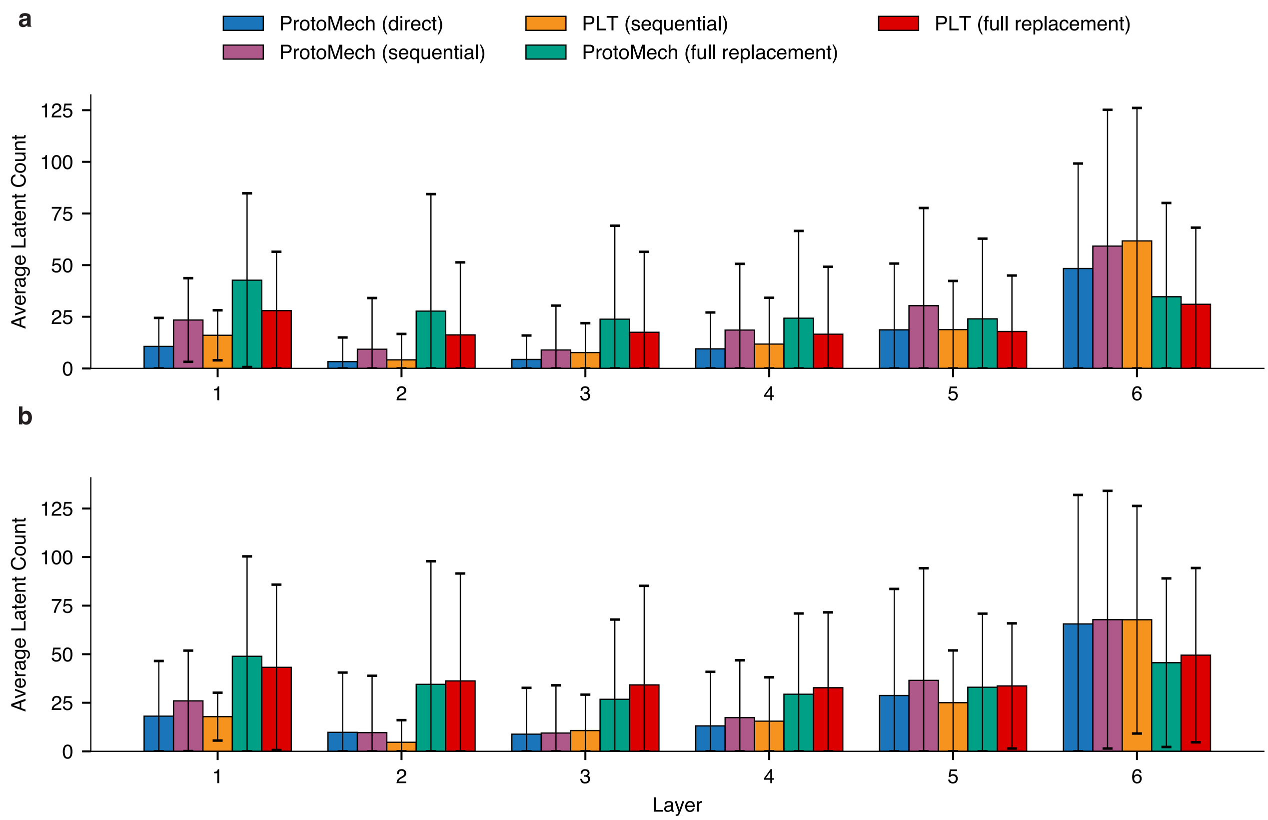

Figure 6. Average number of latents in circuits in all replacement models on **a**, protein family classification and **b**, function prediction tasks.

compounding reconstruction errors. Furthermore, when controlling for the replacement model, ProtoMech consistently outperforms PLT. Lastly, we observe a large gap between the full and sequential replacement models. This phenomenon is consistent with findings in (Ameisen et al., 2025) and, as an aside, we note that developing robust full replacement models remains an active area of research.

Fig. 6 illustrates the average number of latents per layer used in circuit discovery across all replacement models. In ProtoMech, we observe that the latent usage of the direct model is the most sparse, followed by sequential and full replacement. This is unsurprising, as the direct model, is optimized solely to reconstruct $\mathbf{y}^L$ given Equation (2). On the other hand, the full replacement model must retain more latents to support a complete forward pass, requiring information necessary to correctly drive downstream attention mechanisms and maintain signal propagation.

Additionally, we observe a similar phenomenon to (Adams et al., 2025), where the majority of latents being utilized exist are concentrated in the final layers. We attribute this to the information captured at each layer. In the last layer, we observe that latents typically capture granular, amino acid-level biochemical patterns, necessitating a high volume of specific features. In contrast, the middle layers appear to encode higher-level structural motifs, which can be represented more efficiently with a sparser set of latents.

Finally, Fig. 7 details the performance breakdown of our function prediction results. The relative performance of our replacement models remains consistent with Fig. 5.

E.2. Denoising ESM2 in Poor Performance Regimes

We observe that ProtoMech outperforms the original model when ESM2 performs poorly. Fig. 8 details our performance results when ESM2 achieves an F1 score of less than 0.5. Notably, ProtoMech tends to *outperform* the original model on average. Using all latents in the sequential replacement model, ProtoMech attains an F1 of 0.40 ± 0.18 compared to ESM2's

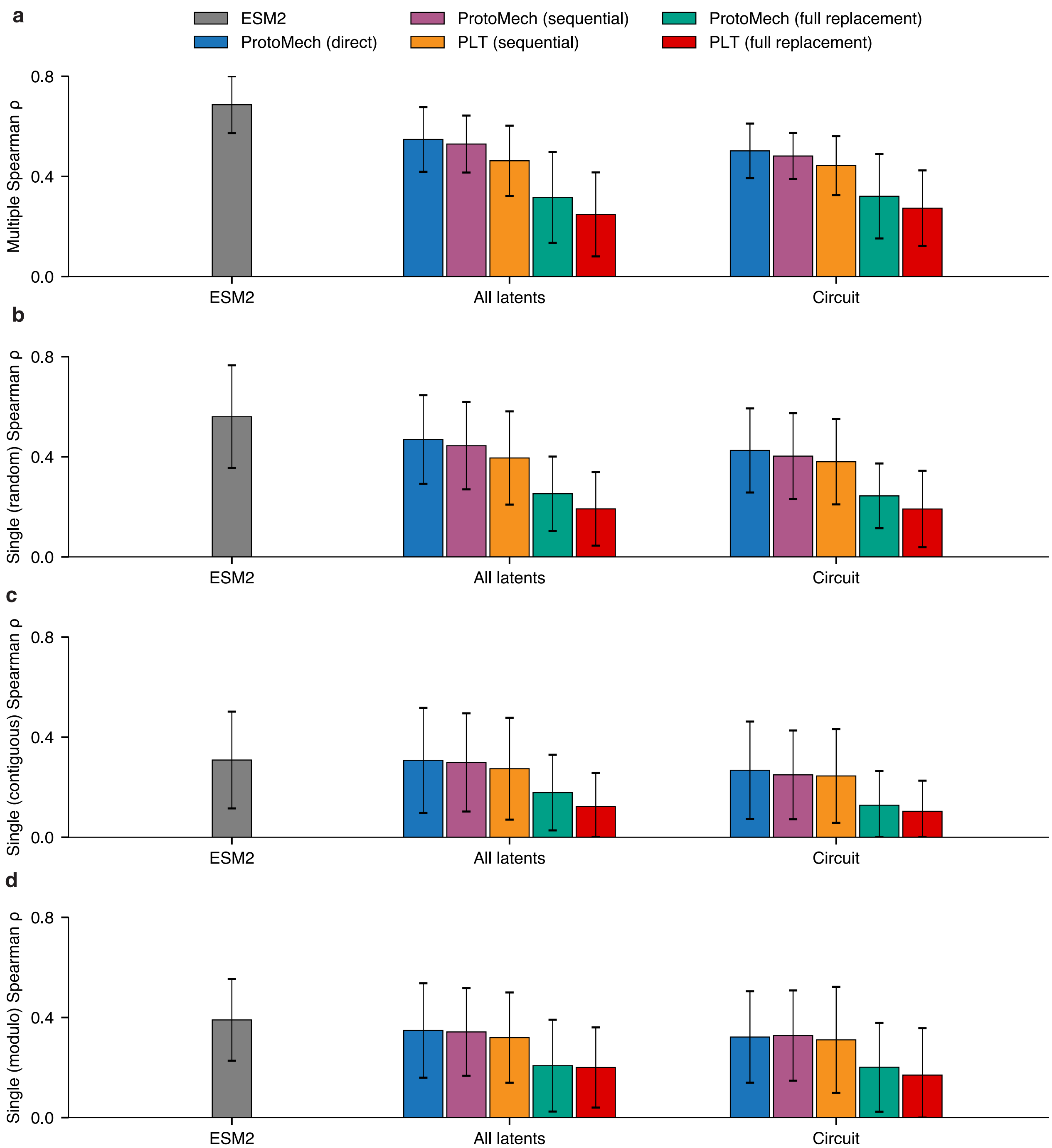

Figure 7. Performance of all replacement models on **a**, mutliple, **b**, single (random), **c**, single (contiguous), and **d**, single (modulo), function prediction tasks.

original performance of 0.39 ± 0.04. In circuit discovery, ProtoMech attains an F1 of 0.43 ± 0.27. Additionally, in function prediction, while ESM2 attains a higher average performance when its Spearman correlation is below 0.2, we find that 37% of circuits are outperforming it. We attribute this gain to ProtoMech effectively denoising ESM2's computation in regimes where the original model performs poorly.

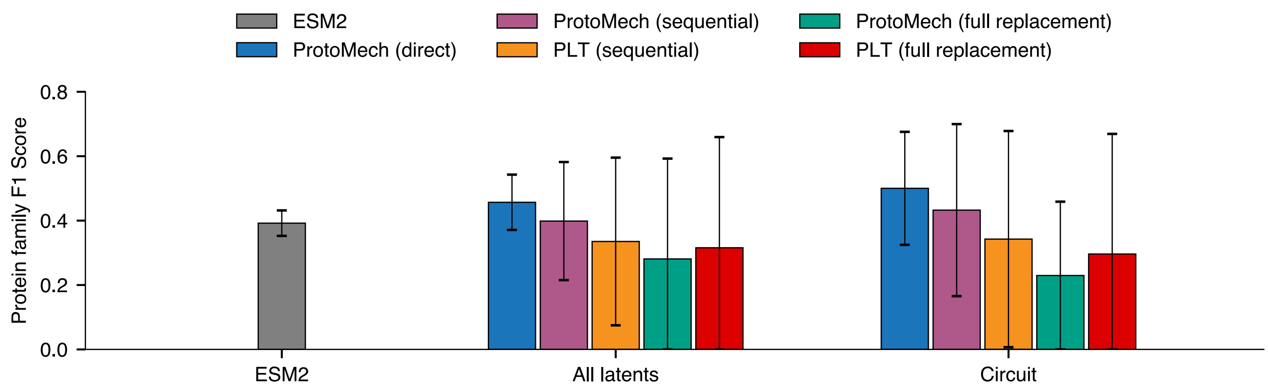

Figure 8. Performance of all replacement models on protein family classification where ESM2 achieved an F1 score of less than 0.5.

E.3. Steering

To determine the optimal intervention strategy for circuit steering, we conducted an ablation study comparing different replacement model architectures. To establish the PLT baseline, we first compared the performance of PLT circuits using the sequential and full replacement models (Table 3). We observe that the sequential replacement model yields superior fitness outcomes, outperforming the full replacement model in 57% of the evaluated cases. Hence, we utilize the sequential replacement model as our PLT baseline in Table 1.

Table 3. Steering results using PLT (sequential) and PLT (full replacement). All variants were constrained to a maximum of five mutations away from the wildtype.

Method	DMS	Mean score ↑	Max score ↑	Top 10% score ↑	Top 20% score ↑
PLT (sequential)	SPG1_STRSG_Olson_2014	**1.97** $\pm$ **0.48**	**3.18**	**2.88** $\pm$ **0.20**	**2.67** $\pm$ **0.25**
	HIS7_YEAST_Pokusaeva_2019	**1.27** $\pm$ **0.06**	1.41	1.38 $\pm$ 0.02	**1.36** $\pm$ **0.03**
	GRB2_HUMAN_Faure_2021	**-0.18** $\pm$ **0.14**	**0.15**	**0.12** $\pm$ **0.05**	**0.03** $\pm$ **0.09**
	GFP_AEQVI_Sarkisyan_2016	4.40 $\pm$ 0.22	5.15	4.92 $\pm$ 0.19	4.76 $\pm$ 0.21
	CAPSD_AAV2S_Sinai_2021	**0.81** $\pm$ **0.53**	**2.05**	**1.87** $\pm$ **0.11**	**1.67** $\pm$ **0.22**
	RASK_HUMAN_Weng_2022_abundance	-0.19 $\pm$ 0.08	**0.06**	**-0.02** $\pm$ **0.06**	**-0.07** $\pm$ **0.07**
	A4_HUMAN_Seuma_2022	-0.05 $\pm$ 0.38	1.03	0.67 $\pm$ 0.24	0.52 $\pm$ 0.24
PLT (full replacement)	SPG1_STRSG_Olson_2014	1.40 $\pm$ 0.61	**3.18**	2.57 $\pm$ 0.33	2.30 $\pm$ 0.37
	HIS7_YEAST_Pokusaeva_2019	1.16 $\pm$ 0.12	**1.52**	**1.40** $\pm$ **0.07**	1.34 $\pm$ 0.08
	GRB2_HUMAN_Faure_2021	**-0.18** $\pm$ **0.13**	0.12	0.09 $\pm$ 0.02	**0.03** $\pm$ **0.08**
	GFP_AEQVI_Sarkisyan_2016	**4.52** $\pm$ **0.25**	**5.16**	**5.00** $\pm$ **0.11**	**4.89** $\pm$ **0.14**
	CAPSD_AAV2S_Sinai_2021	0.77 $\pm$ 0.41	1.82	1.68 $\pm$ 0.08	1.47 $\pm$ 0.23
	RASK_HUMAN_Weng_2022_abundance	**-0.17** $\pm$ **0.08**	-0.01	-0.05 $\pm$ 0.02	**-0.07** $\pm$ **0.03**
	A4_HUMAN_Seuma_2022	**0.40** $\pm$ **0.37**	**1.51**	**1.12** $\pm$ **0.22**	**0.94** $\pm$ **0.24**

With the PLT baseline established, we evaluated the direct, sequential, and full replacement models in ProtoMech. We benchmarked each configuration against the PLT sequential baseline, CAA, and selecting random mutations. We additionally note that, as stated in the Appendix D.3, the number of mutations for the random baseline is based on the number of mutations made in the respective ProtoMech replacement models. As such, the random baseline is expected to fluctuate depending on the ProtoMech replacement model being evaluated. Our results indicate that ProtoMech consistently outperforms the PLT baseline across all configurations, although the margin of improvement varies. Specifically, the ProtoMech direct, sequential, and full replacement models outperformed the PLT baseline in 71% (Table 1), 61% (Table 4), and 57% (Table 5) of cases, respectively, suggesting that the direct model is most suitable for steering ProtoMech.

We use one of the high-fitness proteins designed by the direct replacement model over GB1 to display in Fig. 2. The structure of the variant was generated via AlphaFold3 (Abramson et al., 2024).